

1. Objectives:

1. To understand the concept of Inheritance in Java.
2. To understand the work of 'extend' keyword in Java.
3. To understand the use of 'super' keyword in Java.

2. Introduction:

Inheritance: When a derived class or child class inherits all the members and methods from a base class or parent class is called inheritance. It allows code reuse and helps in creating a hierarchical relationship between classes. The 'extends' keyword is used to inherit the variables and methods of a parent class into a child class. Once a class extends another class, it can access all public and protected members of the parent class.

The 'super' keyword is used in the child class to refer to the parent class. It is commonly used to call the parent class constructor and to access parent class variables and methods. When a constructor is called in a child class, the super() method can be used to initialize the parent class properties.

3. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** Visual Studio Code 1.109

4. Experiment Implementation:

Experiment 1: Practice One with Circle Class

```
package PracticeOne;

public class Box {
    int length;
    int width;
    int height;
    int volume;
    public Box(int l, int w, int h) {
        this.length = l;
        this.width = w;
        this.height = h;
    }
    public int Volume() {
        volume = length * width * height;
        return volume;
    }
    public void show() {
        System.out.println("Volume: " + Volume());
    }
}

package PracticeOne;

public class Circle {

    float radius;
    public Circle(float r) {
        this.radius = r;
    }
    public float Area() {
        System.out.println("Radius of circle: " + (3.14f * radius * radius));
        return (3.14f * radius * radius);
    }
}

package PracticeOne;

public class Main {
    public static void main(String[] args) {
        Box b = new Box(2, 3, 4);
```

```

        b.show();

        Circle c = new Circle(5);
        c.Area();
    }
}

```

Experiment 1 output:

```

● PS D:\BAUST CSE\Java Programing\6th_Lab> & 'C:\Program Files\Java\j
ezwan\AppData\Roaming\Code\User\workspaceStorage\c0035c9f7e02473a05c
Volume: 24
Radius of circle: 78.5
○ PS D:\BAUST CSE\Java Programing\6th_Lab>

```

Experiment 2: Teacher & Student

```

package ExpOne;

public class Information {

    public int id;
    public String name;
    public String gender;
    public void show1() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Gender: " + gender);
    }
}

package ExpOne;

public class Teacher extends Information {

    public String department;
    public void show2() {
        super.show1();
        System.out.println("Department: " + department);
    }
}

package ExpOne;

public class MainClass {
    public static void main(String[] args) {

        Teacher teacher = new Teacher();
        teacher.id = 1;
        teacher.name = "Nakib Sir";
        teacher.gender = "Male";
        teacher.department = "CSE";
        teacher.show2();
    }
}

```

Experiment 2 output:

```

● PS D:\BAUST CSE\Java Programing\6th_Lab> & 'C:\Program Files\Java\j
ezwan\AppData\Roaming\Code\User\workspaceStorage\c0035c9f7e02473a05c
ID: 1
Name: Nakib Sir
Gender: Male
Department: CSE
○ PS D:\BAUST CSE\Java Programing\6th_Lab>

```

Experiment 3: Manager & Employee Class

```

package ExpTwo;

public class Employee {

    public int id;
    public String name;
    public Double salary;
    public Employee(int i, String n, Double s) {
        id = i;
        name = n;
        salary = s;
    }
    public void showInfo() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
    }
}

package ExpTwo;

public class Manager extends Employee {

    public Double bonus;
    public Manager(int i, String n, Double s, Double b) {
        super(i, n, s);
        bonus = b;
    }
    public void showTotalSalary() {
        super.showInfo();
        System.out.println("Bonus: " + bonus);
    }
}

package ExpTwo;

public class Main {

    public static void main(String[] args) {
        Manager m1 = new Manager(1, "Rahim Mama", 5000.00, 10000.00);
        m1.showTotalSalary();
    }
}

```

Experiment 3 output:

```

● PS D:\BAUST CSE\Java Programing\6th_Lab> & 'C:\Program Files\Java\
ezwan\AppData\Roaming\Code\User\workspaceStorage\c0035c9f7e02473a05
ID: 1
Name: Rahim Mama
Salary: 5000.0
Bonus: 10000.0
○ PS D:\BAUST CSE\Java Programing\6th_Lab> █

```

5. Discussion & Conclusion:

In this lab, we learned inheritance using Java. The relationship between parent and child classes was established using the `extends` keyword, which allowed code reuse and reduced redundancy. The `super` keyword was used to access parent class constructors and variables, ensuring proper initialization of inherited properties. Through this lab, the understanding of inheritance, class relationships, and constructor chaining was improved. These concepts are essential for building efficient and scalable object-oriented programs in Java.

7. Reference:

1. <https://www.geeksforgeeks.org/search/?gq=inhheritance> accessed at 6.20 PM on 30th April 2026.